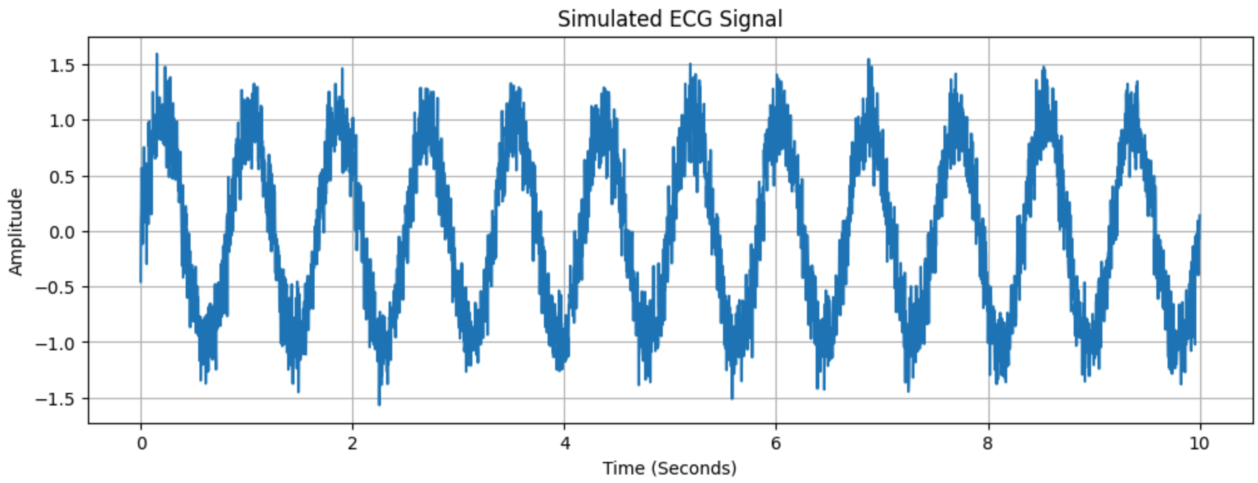
```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal
```

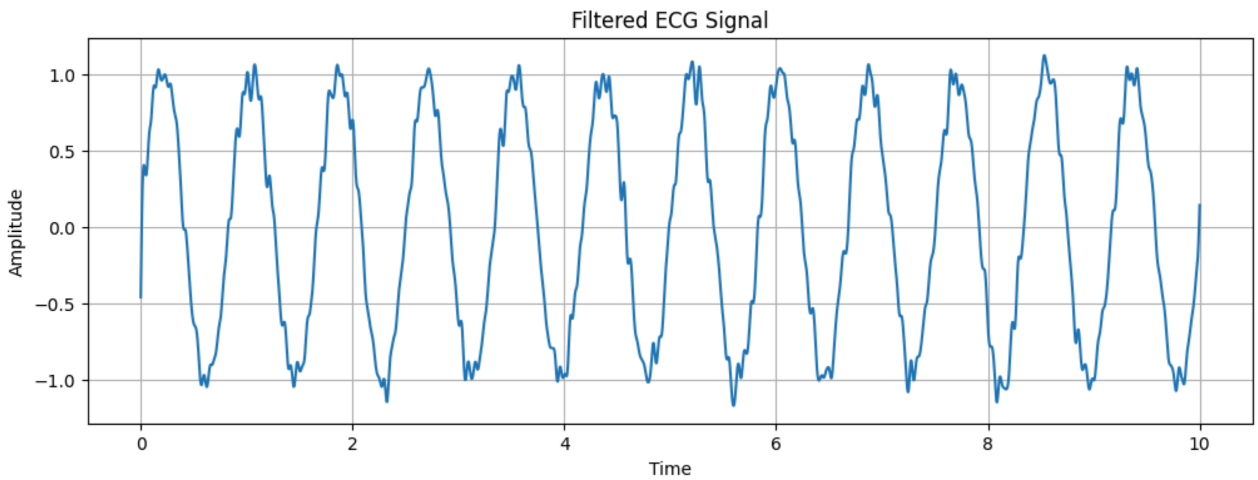
```
fs = 360
t = np.linspace(0, 10, fs * 10)

ecg = np.sin(2 * np.pi * 1.2 * t)
ecg = ecg + 0.2 * np.random.randn(len(t))
```

```
plt.figure(figsize=(12,4))
plt.plot(t, ecg)
plt.title("Simulated ECG Signal")
plt.xlabel("Time (Seconds)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```



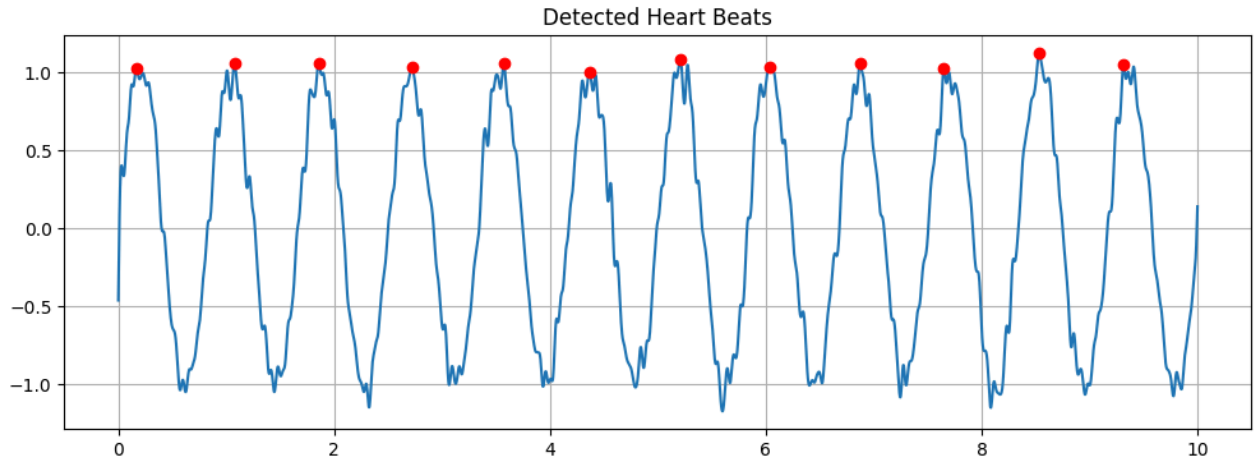

```
b, a = signal.butter(4, 20/(fs/2), 'low')

filtered = signal.filtfilt(b, a, ecg)
```

```
plt.figure(figsize=(12,4))
plt.plot(t, filtered)
plt.title("Filtered ECG Signal")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```




```
peaks, _ = signal.find_peaks(filtered, distance=200)
```

```
plt.figure(figsize=(12,4))

plt.plot(t, filtered)

plt.plot(t[peaks], filtered[peaks], "ro")

plt.title("Detected Heart Beats")

plt.grid(True)

plt.show()
```




```
heart_rate = len(peaks) * 6

print("Estimated Heart Rate =", heart_rate, "BPM")
```

```
Estimated Heart Rate = 72 BPM
```

Start coding or generate with AI.